

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using content-based filtering approach	
Experiment No: 11	Date:02-04-2026	Enrolment No:92301733024

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.

In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:

In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.

Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:

The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

Advantages and Disadvantages:

- **Advantages:**
 - No need for data on other users when applying to similar users.
 - Able to recommend to users with unique tastes.
 - Able to recommend new & popular items

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- Explanations for recommended items.
- **Disadvantages:**
 - Finding the appropriate feature is hard.
 - Doesn't recommend items outside the user profile.

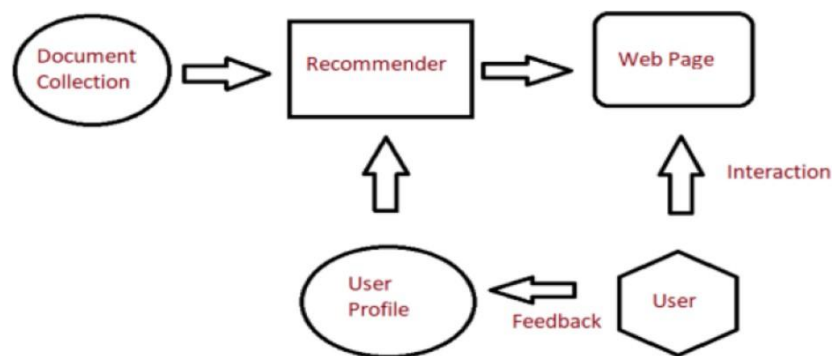


Fig: Recommender System

Pre Lab Exercise:

1. When can you use content-based recommendation system approach?

Content-based recommendation systems are used when we want to recommend items based on a user's past preferences and item features.

2. Write advantages of content-based filtering approach

No need for data from other users

Works well for users with unique tastes

Can recommend new or unpopular items

Provides explainable recommendations

No cold start problem for items

3. Write disadvantages of content-based filtering approach

Difficult to extract meaningful features

Limited to recommending similar items only

Cannot suggest completely new types of items

Requires good quality item descriptions

User profile may become too narrow over time

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Program (Code):

```

import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

df=pd.read_csv("/content/movies.csv")
df.head()

#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']

#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])

cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)

#dropping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()

def recommendation_movie(title,topn=10):
    if title not in indices:
        return "Movie is not found!"
    idx=indices[title]
    sim_scores=list(enumerate(cosine_sim[idx]))
    sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
    sim_scores=sim_scores[1:topn+1]
    movie_indices=[i[0] for i in sim_scores]
    return df["original_title"].iloc[movie_indices]

recommendation_movie("Inception")

```

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original_title	
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance
1930	Stone Cold
914	Central Intelligence
322	The Fifth Element
1271	Pandorum
1267	Duplex

Observation:

The system recommends movies based on similarity of genres and overview using TF-IDF and cosine similarity. Movies with similar themes, storylines, or genres are recommended. However, the recommendations are limited to content similarity and may lack diversity.

Post Lab Exercise:

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.

- Adding more features like cast, director, ratings, keywords
- Combining collaborative + content-based filtering
- Improving preprocessing (stopwords, stemming, lemmatization)
- Adding user feedback to refine recommendations
- Using weighted features instead of simple concatenation

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2. Recommend top-10 “series” using IMDB movie dataset

(Link: <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>). Attach the code with output. Also, write your observation for the same.

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/imdb_top_1000.csv")
df.head()
```

Series_Title	Released_Year	Certificate	Runtime	Genre	IMDB_Rating	Overview	Meta_score	Director	Star1	Star2	Star3	Star4	Star5
The Shawshank Redemption	1994	A	142 min	Drama	9.3	Two imprisoned men bond over a number of years...	80.0	Frank Darabont	Tim Robbins	Morgan Freeman	Bob Gunton	William Sadler	
The Godfather	1972	A	175 min	Crime, Drama	9.2	An organized crime dynasty's aging patriarch t...	100.0	Francis Ford Coppola	Marlon Brando	Al Pacino	James Caan	Diane Keaton	
The Dark Knight	2008	UA	152 min	Action, Crime, Drama	9.0	When the menace known as the Joker wreaks havoc...	84.0	Christopher Nolan	Christian Bale	Heath Ledger	Aaron Eckhart	Michael Caine	
The Godfather: Part II	1974	A	202 min	Crime, Drama	9.0	The early life and career of Vito Corleone in ...	90.0	Francis Ford Coppola	Al Pacino	Robert De Niro	Robert Duvall	Diane Keaton	
12 Angry Men	1957	U	96 min	Crime, Drama	9.0	A jury holdout attempts to prevent a miscarriage of justice by convincing his swing...	96.0	Sidney Lumet	Henry Fonda	Lee J. Cobb	Martin DonOV	John Fiedler	

#preprocessing step

```
df["Genre"]=df["Genre"].fillna("")
df["Overview"]=df["Overview"].fillna("")
df["Content"]=df["Genre"]+" "+df['Overview']
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["Content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
```

#dropping the duplicates

```
indices=pd.Series(df.index,index=df["Series_Title"]).drop_duplicates()
def recommendation_movie(title,topn=10):
    if title not in indices:
```

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```

return "Movie is not found!"
idx=indices[title]
sim_scores=list(enumerate(cosine_sim[idx]))
sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
sim_scores=sim_scores[1:topn+1]
movie_indices=[i[0] for i in sim_scores]
return df["Series_Title"].iloc[movie_indices]
recommendation_movie("Inception")

```

	Series_Title
654	Gattaca
353	Nefes: Vatan Sagolsun
708	À bout de souffle
645	Wo hu cang long
412	Kagemusha
855	The Odd Couple
238	Munna Bhai M.B.B.S.
884	The Peanut Butter Falcon
893	Sicario
677	Predator

Observation:

The system recommends movies/series with similar genres and storyline patterns. The recommendations are relevant but limited to content similarity and do not consider user ratings or popularity.